

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Week 6 Project Summary

By
Ogheneughwe Godwin AKISE

Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 6 Submission – Model Improvement, Error Analysis & Validation

1. What you planned to accomplish

The Week 6 plan was to move beyond the Week 5 Logistic Regression baseline by: analysing the baseline's weaknesses and prediction errors; conducting a structured error analysis of false positives and false negatives; refining features where the error analysis justified it; developing at least one improved modelling approach; fairly comparing the improved model(s) against the baseline; incorporating validated findings from the Data Analytics track; assessing model limitations and risks; and recommending a single candidate model, with defined testing requirements, to carry into Week 7.

2. What you completed

- Reproduced the exact Week 5 baseline (Logistic Regression) on the preserved holdout test set, with reproducibility validated via explicit assertion checks.
- Conducted a full error analysis: confusion-matrix breakdown, descriptive comparison of false negatives vs. true positives and false positives vs. true negatives, and confidence-level (predicted-probability) analysis of the most confidently incorrect predictions.
- Engineered three interaction features (`history_x_lead_time`, `distance_x_lead_time`, `history_interaction`) directly motivated by the error analysis and by Data Analytics track findings.
- Trained and evaluated three candidate improvements — a Refined Logistic Regression, a Random Forest, and a Histogram Gradient Boosting classifier — against the baseline, using a consistent metric suite.
- Ran an overfitting/generalisation check (train vs. test performance) for every model.
- Interpreted the leading model via coefficient analysis and permutation feature importance, and cross-validated these against Data Analytics track findings.
- Built a leakage-safe, out-of-fold threshold-optimisation workflow, selected a provisional, business-aligned operating threshold on training data only, and evaluated it once on the untouched holdout test set.
- Selected a final Week 6 candidate model and defined six concrete Week 7 testing workstreams.
- Produced the Week 6 Model Improvement & Validation Notebook/Report documenting all of the above, including a documented, verifiable cross-track integration evidence artefact.

3. What you improved from Week 5

The Refined Logistic Regression model improved on the Week 5 baseline on both headline ranking metrics: ROC-AUC rose from 0.6873 to 0.6890, and PR-AUC rose from 0.6680 to 0.6717. The improvement is modest but genuine, and — critically — it was achieved without introducing overfitting (the train–test performance gap for the refined model remained as small as the baseline's). Week 6 also delivered something Week 5 could not: a quantified, evidence-based operating threshold. Moving from the default 0.50 threshold to a HealthConnect-aligned 0.45 threshold, selected using training-only evidence, avoided 43 missed no-shows on the holdout test set at a measured cost of 49 additional false-positive contacts. Beyond the headline metrics, Week 6 also produced a documented, evidence-based understanding of where and why the baseline was wrong, confirmation that two untuned tree-based alternatives are not yet ready

(due to overfitting), and a specific, prioritised list of what must be tested before any model is considered for deployment.

4. What you integrated

Two of the three Week 6 engineered features (`history_x_lead_time` and `distance_x_lead_time`) were built directly from interaction directions suggested by the Data Analytics track's Week 5/6 findings, rather than being chosen in isolation. The resulting Refined Logistic Regression coefficients and permutation importance were then cross-checked against the Data Analytics track's descriptive statistics, closing the loop between the two tracks' independent analyses.

5. Which track(s) you collaborated with

Data Analytics Track (primary Week 6 integration partner). The ML Engineering track was also identified as relevant for the downstream pipeline-integration requirements of the eventual candidate model, though the Week 6 integration activity itself was carried out with Data Analytics.

6. What was exchanged

- **Received from Data Analytics:** validated findings on no-show patterns by patient history, booking lead time, distance to clinic, reminder status, appointment type, and appointment timing, plus a specific suggestion to test interaction effects between these variables.
- **Provided to Data Analytics:** the Data Science error-analysis findings, model-comparison results, and the provisional candidate model and threshold requirements for downstream use, so that Data Analytics can target further investigation at the model's current blind spots.
- **Evidence of exchange:** a concrete, auditable artefact (a shared dashboard file) was recorded against this exchange in the Week 6 report, satisfying the brief's requirement for verifiable, not merely narrative, cross-track evidence.

7. What changed as a result

Two of the three interaction features engineered in Week 6 exist specifically because of the Data Analytics track's suggested interaction directions (`lead_time × previous_no_show_history`; `distance × lead_time`). This is a direct, traceable change to the modelling approach — not just a shared narrative — that contributed to the Refined Logistic Regression's improved ROC-AUC and PR-AUC over the baseline.

8. Key findings or development outcomes

- The baseline's errors are not random noise: false negatives and false positives are each driven, in opposite directions, by `booking_lead_days` and `previous_no_shows`, and the model is confidently (not just marginally) wrong in a meaningful share of both error types.
- `booking_lead_days` is by far the single strongest predictor of no-show risk, both by coefficient magnitude and by permutation importance ($\Delta\text{ROC-AUC } 0.159$ — roughly four times the next most important feature).

- Feature engineering produced a real but modest improvement; model-family complexity did not — both tree-based alternatives underperformed Logistic Regression and showed significant overfitting with default hyperparameters.
- The threshold-optimisation work showed that, on the holdout test set, an intervention-efficiency ratio of roughly 1.14 additional false-positive contacts is required for every additional missed no-show avoided when moving from the default to the HealthConnect-aligned threshold — a concrete, business-interpretable figure not previously available.
- Independent validation: the model's own internal feature-importance evidence agrees closely with the Data Analytics track's separately derived descriptive findings, increasing confidence that the model is learning genuine signal, though this is supporting evidence rather than proof of causation.
- The Refined Logistic Regression, evaluated at its provisional HealthConnect-aligned threshold, was retained as the Week 6 candidate model on the basis of its performance, generalisation, interpretability and evidence-based feature engineering.

9. Major challenges

- Balancing two error types with different business costs (missed no-shows vs. unnecessary interventions) without a HealthConnect-confirmed cost weighting to guide threshold selection; this was partly addressed by deriving an internal intervention-efficiency rule, but formal stakeholder confirmation is still required.
- Tree-based models (Random Forest, Histogram Gradient Boosting) showed strong theoretical promise for capturing non-linear interactions but, with default hyperparameters, overfit significantly and could not be fairly assessed against the tuned Logistic Regression baseline within the Week 6 scope.
- The processed Week 5/6 feature matrices do not retain patient identifiers, so the threshold-selection cross-validation could only use ordinary stratified folds rather than the patient-grouped separation used for the original train/test split — a limitation now carried forward as an explicit Week 7 requirement.

10. Important decisions

1. To retain the Week 5 processed data and saved model artefacts as the ground truth for Week 6, rather than reprocessing from raw data — preserving comparability between baseline and improved models.
2. To treat the three new interaction features as testable hypotheses requiring validation through retraining, not as assumed improvements.
3. To select the Refined Logistic Regression, not a tree-based model, as the Week 6 provisional candidate, prioritising demonstrated generalisation and interpretability over unproven complexity.
4. To select the operating threshold using leakage-safe, out-of-fold training evidence only, explicitly freezing it before a single evaluation on the holdout test set, rather than tuning it directly on test data.

11. Remaining issues

- Random Forest and Histogram Gradient Boosting remain untuned; their true potential relative to the Refined Logistic Regression is not yet known.
- No fairness/bias testing across patient sub-groups has yet been performed — required before any production consideration.
- The provisional HealthConnect-aligned threshold (0.45) has not yet been confirmed with HealthConnect stakeholders, since a formal false-positive tolerance or cost weighting has not been supplied.
- Patient identifiers are not currently available in the processed feature matrices, preventing patient-grouped validation of the threshold-selection work; recovering this is an outstanding Week 7 item.

12. Your contribution to the overall HealthConnect project

Week 6 advances the HealthConnect no-show-prediction workstream from an initial benchmark (Week 5) to a validated, modestly improved, well-understood provisional candidate model, with a quantified, leakage-safe operating threshold and a clear, evidence-based rationale for its selection. The documented cross-track integration with Data Analytics, now backed by a concrete evidence artefact, also strengthens the overall HealthConnect project's evidentiary base, since two independent analytical approaches corroborate the same key no-show risk factors — increasing confidence for any HealthConnect stakeholder reviewing the project's findings.

13. Your proposed focus for Week 7

1. Confirm the provisional operating threshold with HealthConnect stakeholders, using the Week 6 threshold-comparison evidence as the starting point rather than re-selecting a threshold from the holdout test set.
2. Perform robust cross-validation — preferably patient-grouped, if patient identifiers can be recovered — to confirm the Refined Logistic Regression's performance and operating threshold are stable across resamples.
3. Test the candidate model for fairness and error rates across patient sub-groups (e.g. gender, age group, distance to clinic).
4. Validate the operational impact of the selected threshold against HealthConnect's actual intervention capacity.
5. Develop a production monitoring plan covering performance tracking, data/concept drift detection, alerting and a retraining schedule.
6. Document deployment and integration requirements: input/output format, inference latency, integration points, and infrastructure dependencies.

References

- AnalystLab Africa. (2026). *HealthConnect Clinic: Data Science baseline modelling report*. Week 5 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Model improvement & validation report*. Week 6 submission.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
<https://doi.org/10.1023/A:1010933404324>
- Dantas, L. F., Fleck, J. L., Cyrino Oliveira, F. L., & Hamacher, S. (2018). No-shows in appointment scheduling – a systematic literature review. *Health Policy*, 122(4), 412–421.
<https://doi.org/10.1016/j.healthpol.2018.02.002>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
<https://scikit-learn.org/stable/modules/preprocessing.html>
- Saeb, S., Lonini, L., Jayaraman, A., Mohr, D. C., & Kording, K. P. (2017). The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), gix019.
<https://doi.org/10.1093/gigascience/gix019>